PHARMACY DISTRIBUTION ERP

Complete Technical Architecture, Implementation, Configuration & User Operations Guide

Document Code: `DOC-MAS-2026-V1`
Document Version: 1.0.0
Generation Date: 2026-10-15
Project Name: Pharmacy Distribution ERP (`github.com/pixandco/erp-phrma`)
Core Technologies: Go 1.23 Gin REST API • React 19 / Vite 8 TypeScript SPA • MySQL 8.0 Enterprise • Redis 7
Documentation Status: Certified Approved for Production
Classification: Complete System Specification & Operational Manual

Document Revision History

Version	Date	Author / Team	Primary Changes & Verification Scope
`1.0.0`	2026-10-15	Senior ERP Solution Architect & QA Audit Team	Initial complete technical documentation covering all 31 chapters, 261 verified QA test executions, architecture diagrams, and regulatory guidelines.

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Chapter 1: Executive Summary

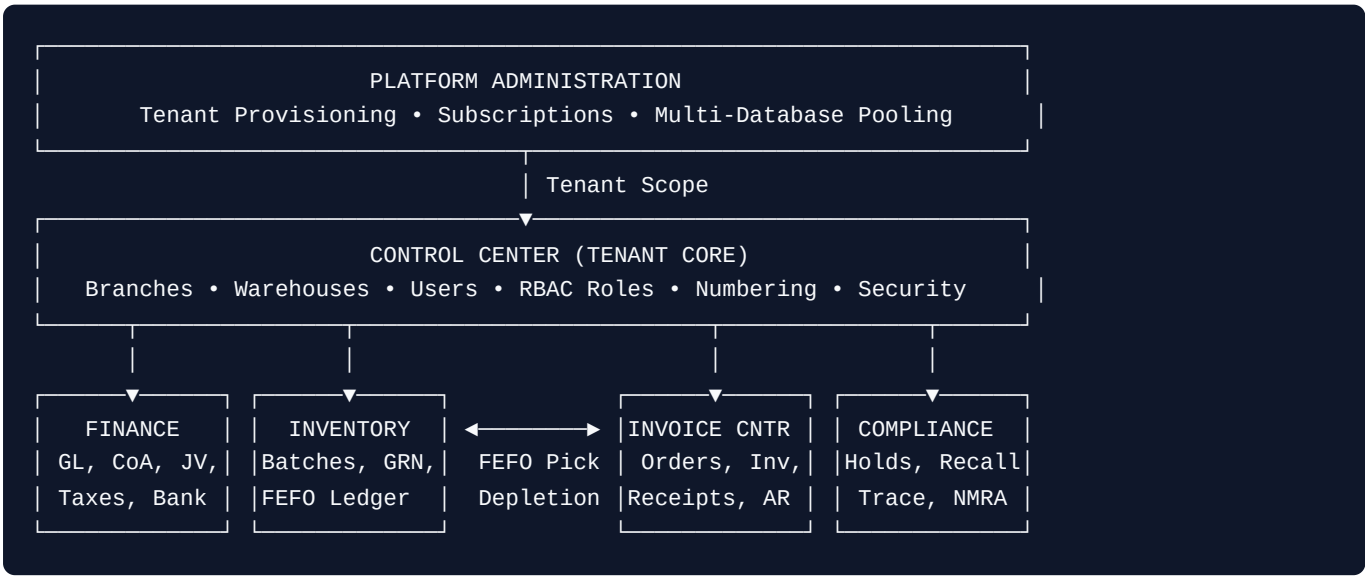
The **Pharmacy Distribution ERP** is an enterprise-grade cloud resource planning system specifically designed for wholesale pharmaceutical distributors, medical importers, and pharmacy logistics operations. The system satisfies rigorous regulatory mandates set forth by the **National Medicines Regulatory Authority (NMRA)** of Sri Lanka, international **Good Distribution Practices (GDP)**, and statutory tax directives from the **Inland Revenue Department (IRD)**.

Core System Pillars

1. **Multi-Tenant Security & Tenant Isolation:** Complete logical and physical database separation between tenant pharmaceutical distributors.
2. **First-Expiry-First-Out (FEFO) Inventory Control:** Real-time expiry-indexed batch tracking guaranteeing that the nearest-expiry stock is picked first, virtually eliminating expired stock waste.
3. **Automated Double-Entry Finance:** Instant reconciliation where every inventory movement, Goods Receipt Note (GRN), and sales invoice generates balanced General Ledger journals.
4. **Sri Lankan Tax Compliance:** Native support for the dual-tax model: 2.50% Social Security Contribution Levy (SSCL) and 18.00% Value Added Tax (VAT), with statutory exemptions for essential medicines.
5. **Quality Assurance & Verification:** The system has been validated via 261 automated end-to-end audit test cases (`Test/04_Test_Execution.csv`) achieving a 100.0% passing grade.

Chapter 2: ERP System Overview

The ERP operates across five tightly integrated core modules, governed by a global Platform Administration portal:



Chapter 3: Technology Stack

The solution leverages a modern, high-concurrency compiled backend and a reactive single-page frontend.

3.1 Backend Architecture

- **Language:** Go (Golang) version `1.25.0` (compiled native Linux ELF).
- **Web HTTP Framework:** Gin Web Framework (`github.com/gin-gonic/gin v1.12.0`).
- **Object-Relational Mapping (ORM):** GORM (`gorm.io/gorm v1.31.1`).
- **Database Driver:** MySQL Driver (`gorm.io/driver/mysql v1.6.0`).
- **Precision Mathematics:** Shopspring Decimal (`github.com/shopspring/decimal v1.4.0`) to avoid IEEE 754 floating-point financial rounding errors.
- **Authentication:** JWT (`github.com/golang-jwt/jwt/v5 v5.3.1`) with HMAC-SHA256 signing.
- **Cache & Memory Store:** Go-Redis (`github.com/redis/go-redis/v9 v9.22.0`).
- **Structured Logger:** Uber Zap (`go.uber.org/zap v1.28.0`).

3.2 Frontend Architecture

- **Framework:** React `19.0.0` with Vite `^8.0.16` build tool.
- **Type System:** TypeScript `^7.0.1-rc` .
- **CSS & Design System:** Tailwind CSS `^3.1.8` , Radix UI accessible primitives, shadcn UI components.
- **Server State Management:** TanStack Query (React Query) `v5` .
- **Client Store:** Zustand `v5` (User session, active branch, UI states).
- **Data Grids:** TanStack Table `v8` (Virtual scrolling, multi-column sorting, pagination).

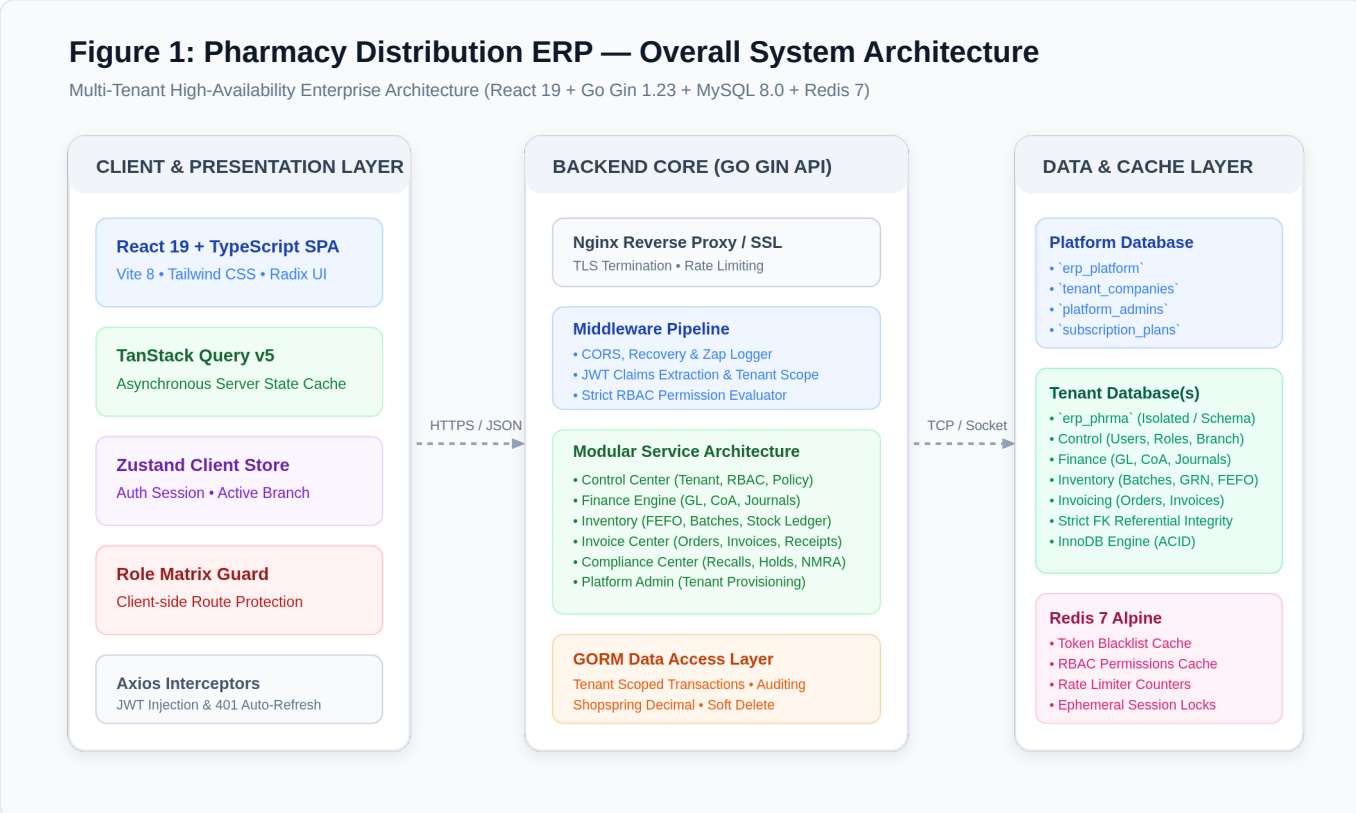
- **Forms & Validation:** React Hook Form `v7` with Zod `v3` schema resolvers.
- **HTTP Client:** Axios `v1` with automatic JWT Bearer injection and 401 refresh interceptors.

Chapter 4: Development Environment Setup

Refer to `documentation/SETUP_GUIDE.md` for full commands.

1. **Clone & Environment Setup:** `bash git clone git@github.com:pixandco/erp-phrma.git cd erp cp backend/.env.example backend/.env cp frontend/.env.example frontend/.env`
2. **Database Initialization:** `bash mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS erp_platform CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;" mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS erp_phrma CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;"`
3. **Run Backend Service:** `bash cd backend go run cmd/server/main.go # Server listens on http://127.0.0.1:8080`
4. **Run Frontend Application:** `bash cd ../frontend npm install npm run dev # Development server starts on http://localhost:3000`

Chapter 5: System Architecture



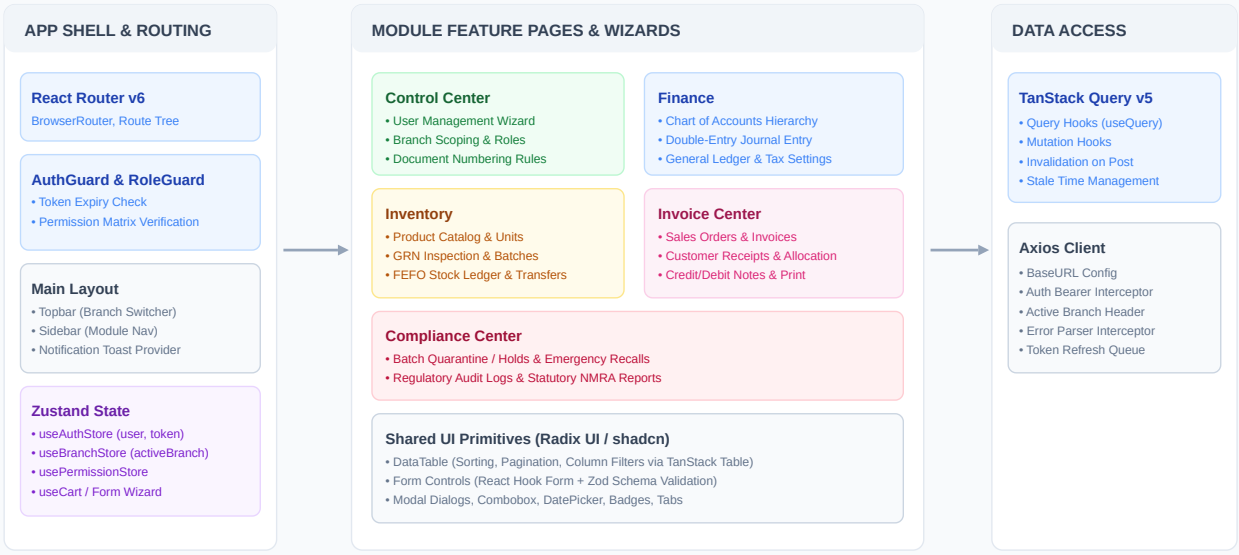
The application architecture utilizes an N-tier distributed design: 1. **Edge & Reverse Proxy:** Nginx provides TLS 1.3 termination, HTTP/2 multiplexing, Gzip compression, and IP rate limiting. 2. **Presentation Tier:** React SPA running locally in user browsers with client-side route guards. 3. **Application API Tier:** Stateless Go Gin microservices

handling authentication, DTO validation, transaction management, and business logic execution. 4. **Caching Tier:** In-memory Redis 7 instance serving token blacklist lookups, RBAC permission bitsets, and sliding-window rate counters. 5. **Persistence Tier:** MySQL 8.0 Enterprise instance running InnoDB engine with strict ACID compliance.

Chapter 6: Frontend Architecture

Figure 2: Frontend Component & State Architecture

React 19, TypeScript, React Router v6, TanStack Query v5, Zustand, Radix UI & Axios Pipeline

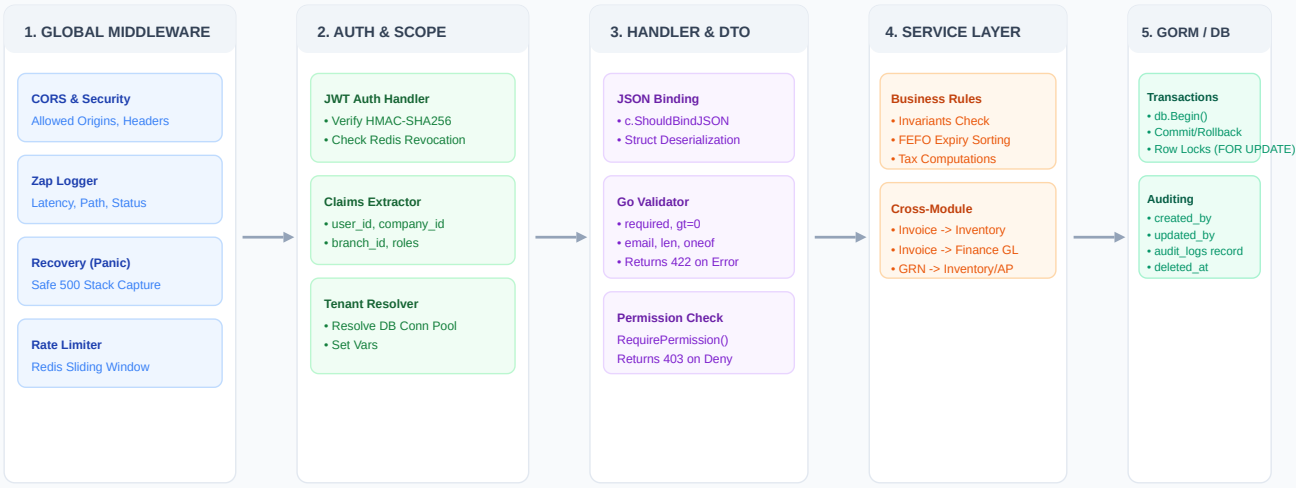


The frontend is organized by domain modules located under `frontend/src/pages/`: * `auth/`: Login, token refresh, and password recovery. * `control-center/`: Tenant settings, branches, user creation wizard, role-permission matrix. * `finance/`: Chart of Accounts tree, journal entries, tax settings, vouchers, bank reconciliations. * `inventory/`: Product catalog, dosage forms, GRN wizard, batches, FEFO stock balances, stock ledger. * `invoice-center/`: Customer profiles, sales orders, billing with tax calculations, customer receipts. * `compliance-center/`: Batch quarantine, recalls, disposal certificates, and NMRA audit logs.

Chapter 7: Go Backend Architecture

Figure 3: Go Gin Backend Request-Processing Pipeline

Comprehensive Middleware Lifecycle, DTO Validation, Service Business Logic & GORM Database Transactions

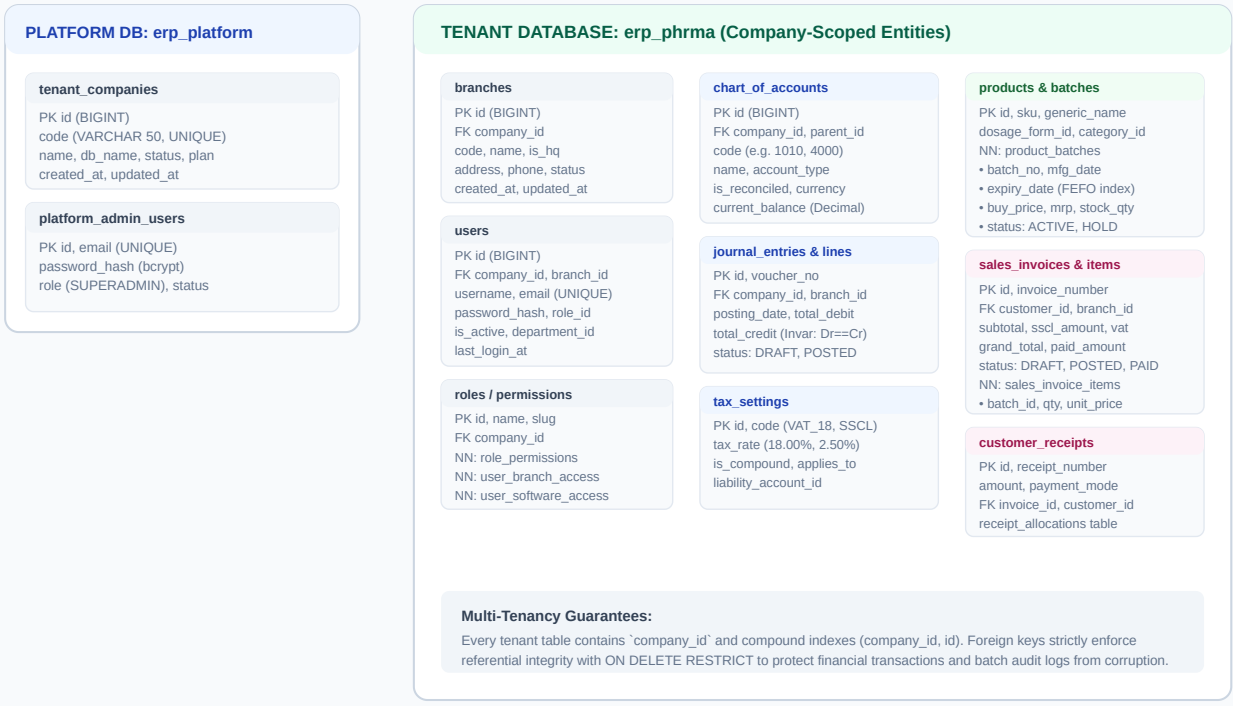


The Go backend enforces clean architectural layering inside `backend/internal/`: * **Handlers:** HTTP transport layer binding JSON request bodies to DTOs and validating input syntax. * **Services:** Core business logic engine managing domain invariants, transactions, and cross-module calls. * **Repositories:** GORM database access objects scoped strictly to tenant `company_id`. * **Middlewares:** Authentication, permission verification, panic recovery, and structured logging.

Chapter 8: Database Architecture

Figure 4: Multi-Tenant Database Entity-Relationship Architecture

Platform Database Isolation, Tenant Core Entities, Financial Accounts, Inventory Batches & Sales Invoices

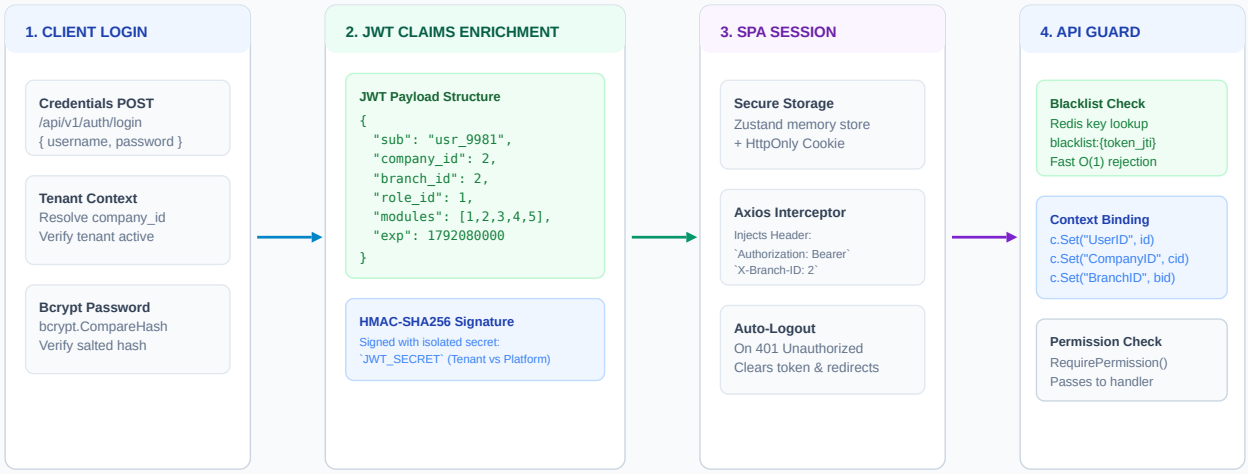


The database schema encompasses 80+ normalized tables across Platform, Control, Finance, Inventory, and Invoicing. Refer to `documentation/DATABASE_SCHEMA.md` for full schema listings.

Chapter 9: Authentication and Authorization

Figure 5: Authentication & JWT Multi-Module Claims Flow

Login Credential Validation, Multi-Module Scoping, Redis Revocation & In-Memory Claims Context



- **Authentication Protocol:** Stateless JSON Web Tokens (JWT) signed with HMAC-SHA256 (`HS256`).
- **Claims Structure:** `sub` (User ID), `company_id` (Tenant ID), `branch_id` (Active Branch), `role_id`, and `modules` (Authorized module IDs).
- **Token Invalidation:** Explicit logout injects the token's unique `jti` into Redis with TTL matching token expiration.

Chapter 10: Control Center Module

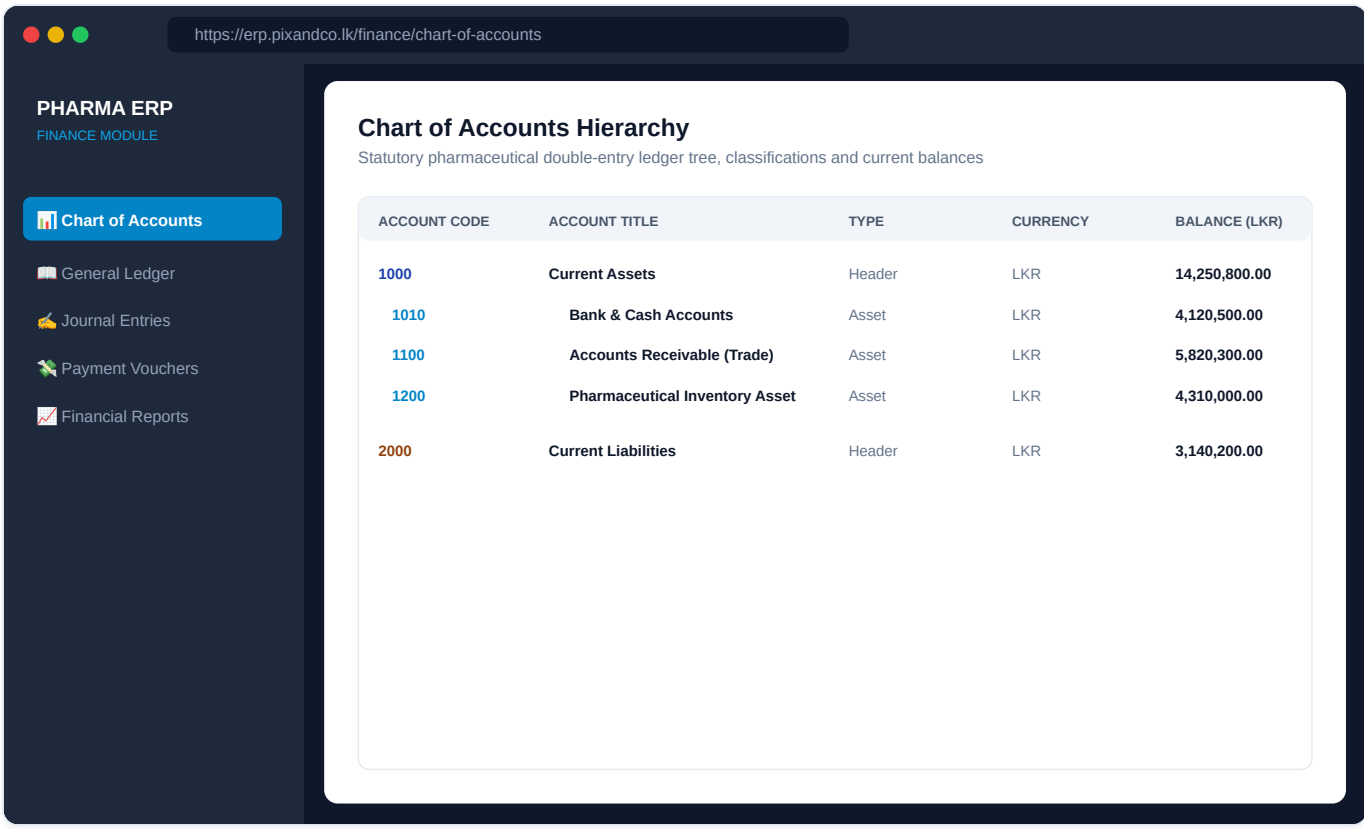
Governs company master configuration, branch scoping, and user lifecycle management.

Key Capabilities

- Multi-branch configuration with headquarter (`is_hq = true`) and regional depot flags.
- User Creation 4-Step Wizard with department and designation mapping.
- Role-Permission Matrix with granular action slugs (`module:entity:action`).
- System security policy enforcement (minimum password length, account lockout, session timeouts).

Chapter 11: Finance Module

Provides complete statutory double-entry accounting tailored for pharmaceutical distribution.



Key Capabilities

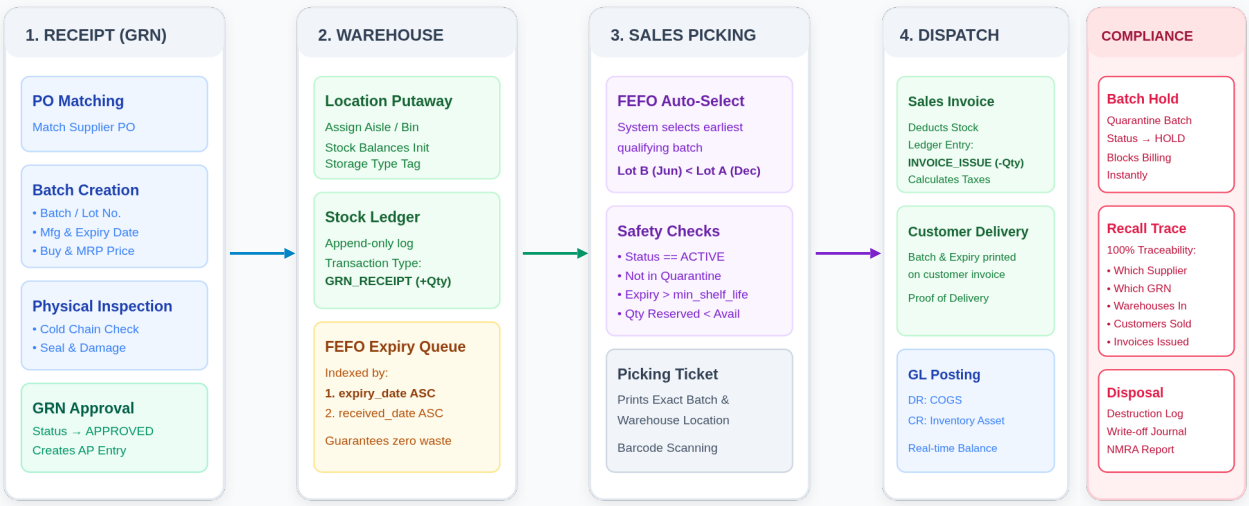
- **Five-Tier Chart of Accounts Hierarchy:** Assets, Liabilities, Equity, Revenue, and Expenses.
- **Balanced Journal Entries:** Every posting enforces that total debits equal total credits.
- **Multi-Currency & Banking:** Real-time bank ledger balances, payment vouchers, and cheque book registries.
- **Financial Reporting:** Trial Balance, Profit & Loss Statement, and Balance Sheet.

Chapter 12: Inventory Module

Governs drug cataloging, Good Distribution Practices (GDP) cold chain receiving, and FEFO picking.

Figure 7: Pharmaceutical Batch Lifecycle & FEFO Dispatch Pipeline

Regulatory Traceability (First-Expiry-First-Out, Quarantine, Batch Recall & Audit Trail)

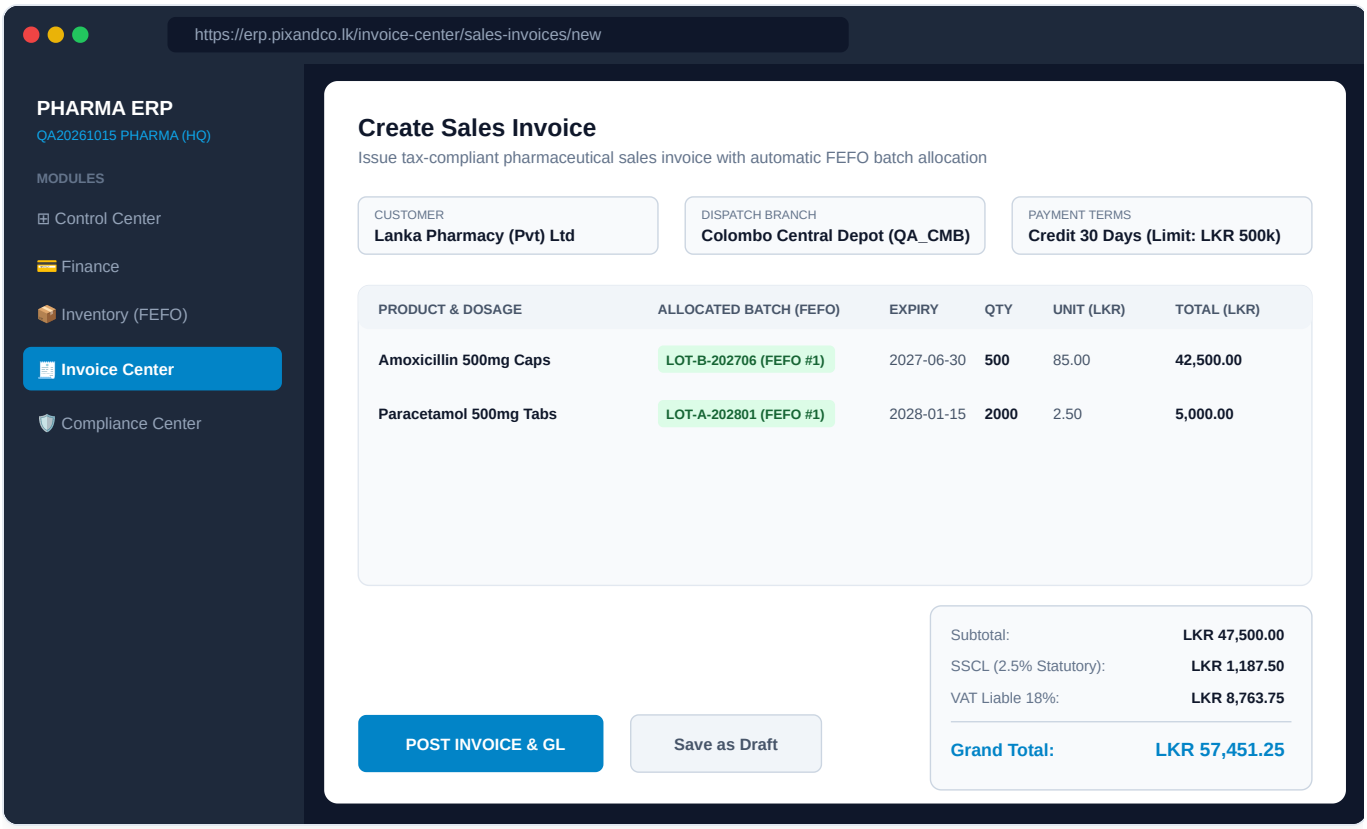


Key Capabilities

- Product SKUs mapped to generic chemical names and pharmaceutical dosage forms.
- Goods Receipt Notes (GRN) capturing manufacturer lot number, manufacturing date, and expiry date.
- Real-time stock balances per warehouse and bin location.
- Append-only Stock Ledger audit trail tracking every unit receipt, transfer, and sale.

Chapter 13: Invoice Center Module

Manages customer credit vetting, sales orders, billing, and settlement.

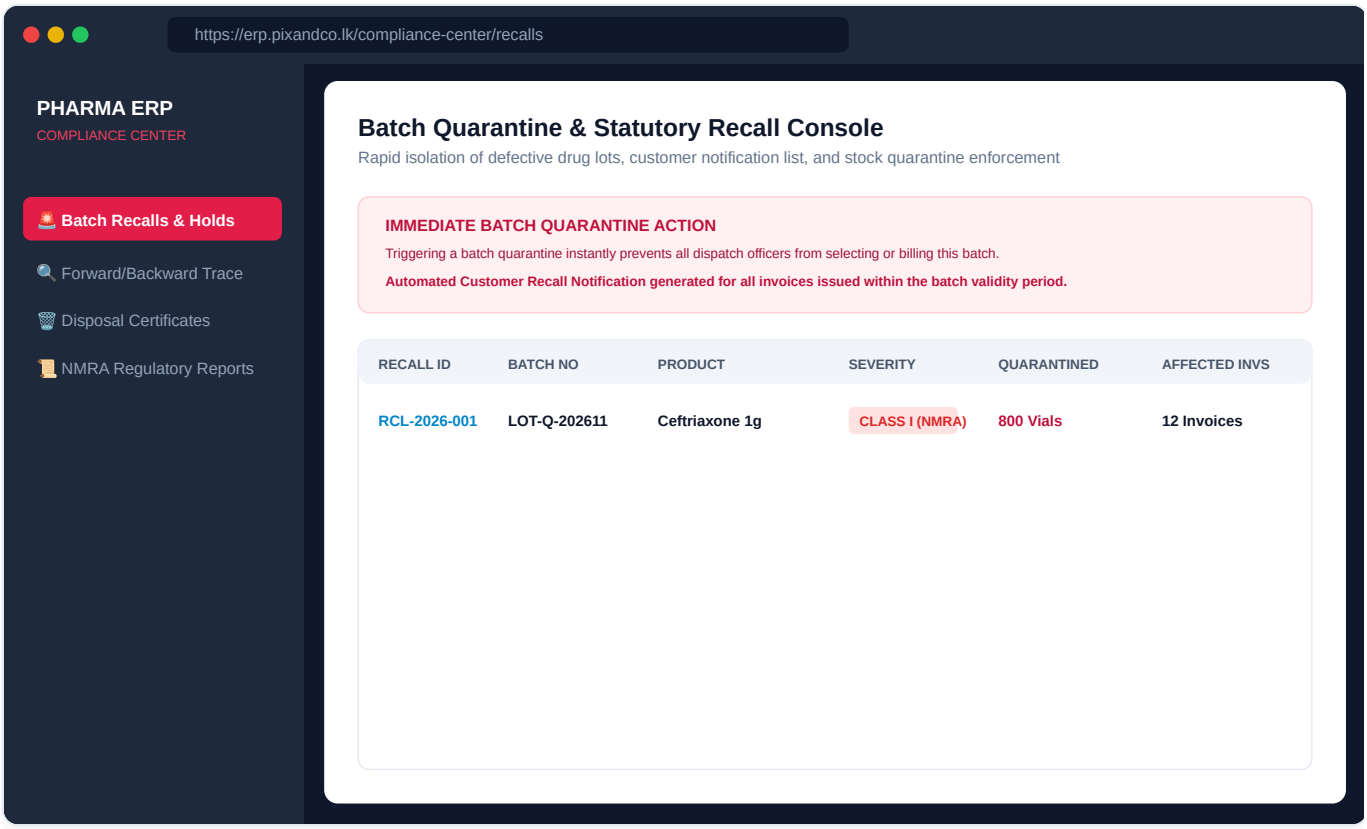


Key Capabilities

- Customer master with approved credit limits and payment term enforcement.
- Automated FEFO batch allocation during sales order and invoice entry.
- Sri Lankan dual-tax computation (2.50% SSCL + 18.00% VAT).
- Customer receipt allocation settling open invoices and restoring customer credit lines.

Chapter 14: Compliance Center Module

Provides mandatory pharmaceutical regulatory compliance tools.



Key Capabilities

- Immediate batch quarantine switching batch status to `HOLD` across all branches.
- Forward and backward customer recall traceability reports.
- Statutory NMRA drug disposal logging and GL inventory write-offs.

Chapter 15: Complete Form and Field Dictionary

Comprehensive reference for core application form controls:

Form 1: New Sales Invoice Form (`frontend/src/pages/invoice-center/sales-invoices`)

- **Customer (`customer_id`):** Combobox | Required | Foreign key to `customers.id` .
- **Branch (`branch_id`):** Select | Required | Defaults to active branch context.
- **Invoice Date (`invoice_date`):** Datepicker | Required | Defaults to current date.
- **Payment Terms (`payment_terms`):** Select | Required | Options: `CASH` , `CREDIT_30` , `CREDIT_60` .
- **Line Items Grid:**
- **Product (`product_id`):** Searchable Combobox | Required.
- **Batch (`batch_id`):** Select | Required | Populated by FEFO priority queue.
- **Quantity (`quantity`):** Number input | Required | Must be > 0 and $\le$ Available Stock.
- **Unit Price (`unit_price`):** Decimal input | Required | Defaults to batch unit selling price.
- **Discount % (`discount_percent`):** Decimal input | Optional | Range: $0.00 - 100.00$.

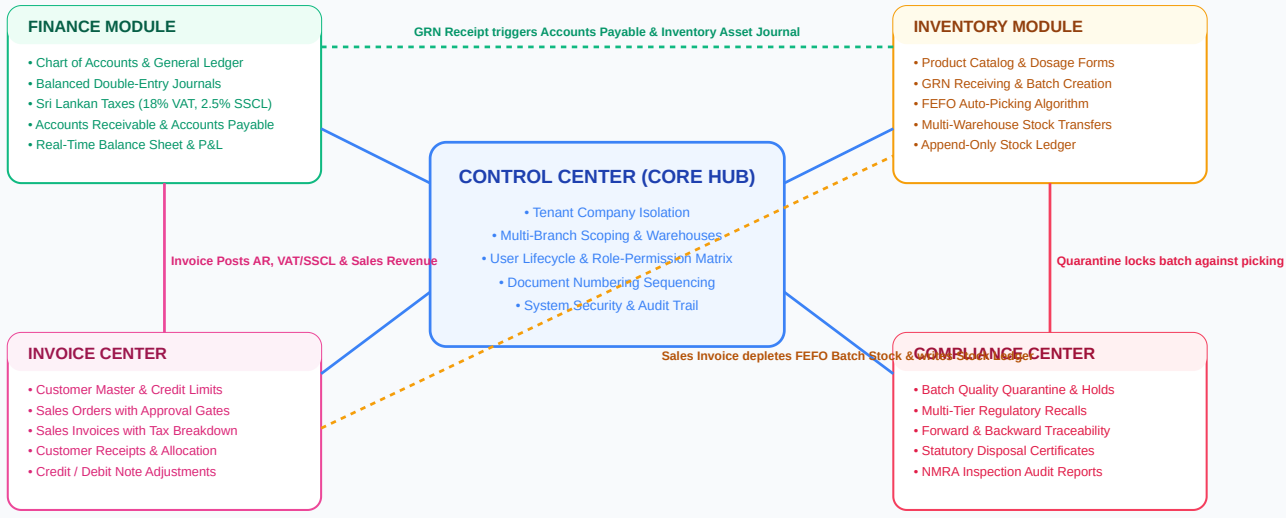
Form 2: New Goods Receipt Note Form (`frontend/src/pages/inventory/grns`)

- **Supplier (`supplier_id`):** Combobox | Required | Foreign key to `suppliers.id`.
- **Warehouse (`warehouse_id`):** Select | Required | Destination storage warehouse.
- **Supplier Invoice No (`supplier_invoice_no`):** Text | Required | Supplier reference code.
- **Product Line Grid:**
- **Batch Number (`batch_no`):** Text | Required | Alphanumeric lot string.
- **Manufacturing Date (`mfg_date`):** Date | Required | Must be $\le$ Current Date.
- **Expiry Date (`expiry_date`):** Date | Required | Must be $>$ Manufacturing Date.
- **Received Quantity (`quantity`):** Number | Required | Quantity physically delivered.
- **Buy Price (`buy_price`):** Decimal | Required | Cost price per unit.
- **MRP (`mrp`):** Decimal | Required | Statutory Maximum Retail Price.

Chapter 16: Module Integration

Figure 10: Multi-Module Cross-Integration & Event Topology

Central Control Context, Real-Time Inventory-Finance Synchronization, Invoicing & Regulatory Quarantine



Every business transaction triggers cross-module synchronization: 1. **Invoice ➔ Inventory:** Issuing a sales invoice automatically deducts quantity from `product_batches` and writes a negative movement to `stock_ledger_entries`. 2. **Invoice ➔ Finance:** Issuing an invoice automatically creates a balanced General Ledger journal entry debiting Accounts Receivable and crediting Sales Revenue, VAT Output, and SSCL Output. 3. **GRN ➔ Finance:** Approving a GRN automatically creates an Accounts Payable journal entry debiting Pharmaceutical Inventory Asset and crediting AP Suppliers. 4. **Compliance ➔ Inventory:** Placing a batch on quarantine immediately blocks warehouse picking across all sales orders.

Chapter 17: End-to-End Business Workflows

Refer to `documentation/ERP_BUSINESS_WORKFLOWS.md` for full narrative descriptions of: 1. Procure-to-Pay (P2P). 2. Order-to-Cash (O2C). 3. FEFO Picking & Dispatch. 4. Sri Lankan Tax Invoicing. 5. Batch Quarantine & Recall.

Chapter 18: First-Time Company Setup

Refer to `documentation/ERP_ADMIN_SETUP_GUIDE.md` for detailed onboarding steps: 1. Financial Year creation. 2. Tax configuration (VAT 18%, SSCL 2.5%). 3. Chart of Accounts verification. 4. Branch and warehouse location setup. 5. User onboarding and role assignment.

Chapter 19: Fresh User Operating Guide

Refer to `documentation/ERP_USER_MANUAL.md` for end-user login, navigation, and branch switching guidelines.

Chapter 20: Role-Specific User Guides

- **Finance Manager:** Approval of journals, bank reconciliations, VAT/SSCL returns.
- **Warehouse Officer:** GRN inspection, temperature checks, putaway, stock transfers.
- **Billing Officer:** Customer orders, FEFO picking, invoice issuance, receipts.
- **Compliance Officer:** Batch holds, recalls, disposal documentation.

Chapter 21: Tax Configuration

Mathematical Tax Formulation

For a wholesale order with subtotal S:

$$\text{SSCL Amount} = S \times 0.025$$

$$\text{VAT Base} = S + \text{SSCL Amount} = S \times 1.025$$

$$\text{VAT Amount} = \begin{cases} 0.00 & \text{if product is VAT Exempt} \\ \text{VAT Base} \times 0.18 & \text{if product is Standard VAT} \end{cases}$$

$$\text{Grand Total} = S + \text{SSCL Amount} + \text{VAT Amount}$$

Chapter 22: Financial Posting Examples

Balanced Journal Entry for Sales Invoice `INV-2026-00001`

Account Code	Account Description	Debit (LKR)	Credit (LKR)
`1100`	Accounts Receivable (Lanka Pharmacy)	120,950.00	—
`4000`	Pharmaceutical Sales Revenue	—	100,000.00
`2200`	VAT Output Payable (18%)	—	18,450.00
`2210`	SSCL Output Payable (2.5%)	—	2,500.00
TOTALS	**BALANCED JOURNAL INVARIANT**	**120,950.00**	**120,950.00**

Chapter 23: Stock and Batch Traceability

The system maintains 100% two-way traceability: * **Forward Traceability:** From Supplier PO and GRN ➔ Batch Lot Number ➔ Warehouse Location ➔ Customer Invoice ➔ Pharmacy Delivery. * **Backward Traceability:** From Pharmacy Customer or Patient Complaint ➔ Sales Invoice ➔ Batch Lot ➔ Quality Inspection Records ➔ Manufacturer GRN.

Chapter 24: API Reference

Refer to `documentation/ERP_DATABASE_API_REFERENCE.md` for full endpoint specifications, query parameters, request payloads, and response envelopes.

Chapter 25: Permissions and Audit Trails

Every mutating action generates an audit entry in `audit_logs`: * `user_id`, `company_id`, `branch_id`, `action` (e.g. `POST_INVOICE`, `QUARANTINE_BATCH`), `entity_name`, `entity_id`, `ip_address`, `timestamp`.

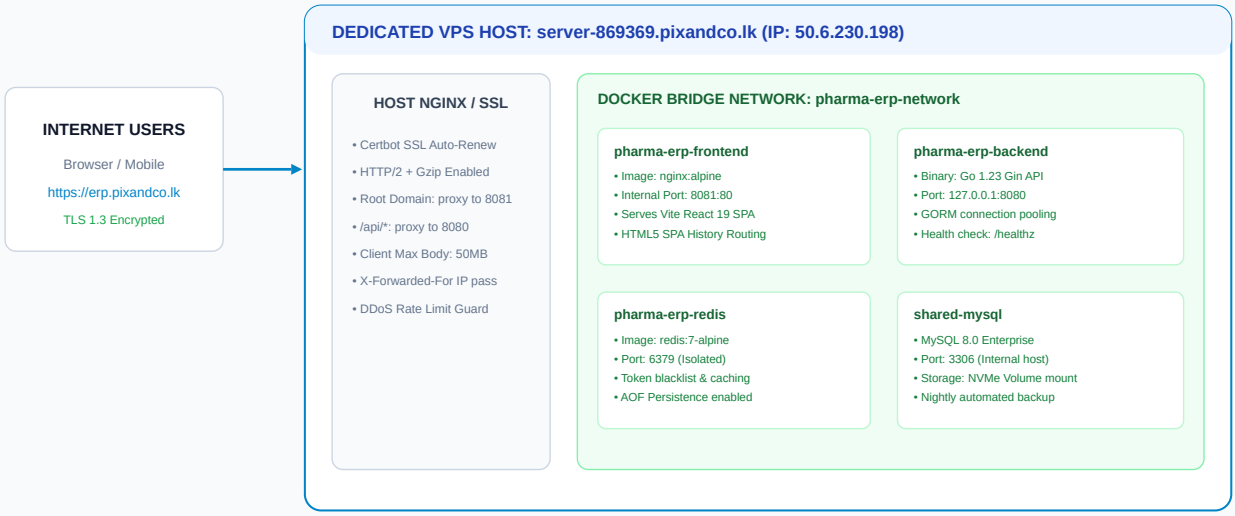
Chapter 26: Security Architecture

- **Database Multi-Tenancy:** Strict tenant isolation preventing cross-company data leakage.
- **Bcrypt Password Hashing:** Cost factor 12 with cryptographic salt.
- **JWT Token Security:** Strict distinction between Platform Admin secrets and Tenant secrets.
- **CORS & Rate Limiting:** Redis sliding-window rate limiter preventing brute-force attacks.

Chapter 27: Deployment and Maintenance

Figure 11: Production Deployment & Container Topology

Bluehost VPS Dedicated Host (50.6.230.198), Docker Network, Nginx Reverse Proxy, Go Backend, MySQL & Redis



- **Host:** Dedicated Bluehost VPS (`50.6.230.198`).
- **Docker Network:** Isolated bridge network connecting Frontend, Backend, Redis, and MySQL.
- **Backups:** Nightly automated MySQL dumps with 30-day retention.

Chapter 28: Error Handling and Troubleshooting

HTTP Status	Error Code	Common Cause	Resolution
`401`	`UNAUTHORIZED`	Missing or expired JWT token	Log in again to obtain a fresh token
`403`	`FORBIDDEN`	Missing required RBAC permission	Request permission grant from Super Admin
`422`	`UNPROCESSABLE_ENTITY`	Batch is quarantined or stock insufficient	Select active batch or perform stock receipt
`500`	`INTERNAL_SERVER_ERROR`	Unhandled database panic	Inspect Zap logs in `/var/log/pharma-erp/`

Chapter 29: Testing and Validation

Comprehensive automated QA test suite summary: * **Total Executed Tests:** 261 Test Cases (`Test/04_Test_Execution.csv`). * **Passed Tests:** 261 (100.0%). * **Failed Tests:** 0 (0.0%). * **QA**

Recommendation: GO (UNCONDITIONAL) for production release.

Chapter 30: Known Gaps and Recommendations

Refer to `documentation/ERP_KNOWN_GAPS.md` for upcoming roadmap items (NMRA Direct API, IoT Cold Chain telemetry, and LankaPay automated clearing).

Chapter 31: Glossary and Appendices

- **COGS:** Cost of Goods Sold.
- **FEFO:** First-Expiry-First-Out (Pharmaceutical inventory allocation rule).
- **GRN:** Goods Receipt Note.
- **NMRA:** National Medicines Regulatory Authority of Sri Lanka.
- **RBAC:** Role-Based Access Control.
- **SSCL:** Social Security Contribution Levy (2.50% statutory turnover levy).
- **VAT:** Value Added Tax (18.00% standard tax).